

Lecture VIII - Pandas and AI Craft

Programming with Python

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Checkpoint 4 — Sessions VI–VII

The first **40 minutes** are the checkpoint. It starts **now**, before the investor opens her data room.

- **Individual work:** no neighbors, no chat
- **AI tools are allowed:** being able to **VERIFY** the output is the skill being graded
- The **link and QR** are handed out in class. Open it and start
- ~6 short tasks: write code, trace code, fix a bug, answer a multiple choice
- It sweeps **Sessions VI–VII:** modules, `random` and seeds, NumPy arrays and masks

...

When you're done: menu → *Download* → *Download Python code* → upload the `.py` to the “**Checkpoint 4**” assignment on Moodle. **No retakes:** one sitting.

...

Note

The green  live checks are **provisional**. The final grading runs on our side. And take a breath: everything in it was rehearsed in the labs.

Episode 8: The Data Room

Show me the data

Pens down. The checkpoint is behind you. This morning the rival chain **MunchCorp** put out a press release bragging about their “**data-driven growth**.” The evidence attached: exactly **one pie chart**. The investor slid it across the table and said, “*Yours will be better. Show me the data.*”

...

Kevin didn't wait. He'd pasted the question into a chatbot, proudly turned the laptop around, and hit run:

```
AttributeError: 'DataFrame' object has no attribute 'overview'
```

...

The AI sounded certain. The method it called **does not exist**. Today is about that difference, and about the tool that answers the investor for real: **pandas**.

AI joins the team — professionally

Prompting that works: context + constraints

An AI is only as good as what you tell it. Two things turn a vague request into a useful answer:

- **Context** — say what the data is: “I have a DataFrame `orders` with columns `zone` (text) and `total_eur` (float).”
- **Constraints** — say exactly what you want *back*: “Give me the total `total_eur` for zone `Nord`, as a single number.”

...

“I have a pandas DataFrame `orders` with a text column `zone` and a float column `total_eur`. Write one line that returns the total `total_eur` for rows where `zone` is `Nord`.”

Vague in, vague out. Specific in, checkable out, and then you **iterate**.

The verify workflow

Kevin’s mistake wasn’t *using* AI. It was **shipping without checking**. Every line an AI hands you gets three passes:

1. **Read it**: do you understand what each line claims to do?
2. **Run it**: does it actually execute, or does it crash?
3. **Test it**: try it on a small case where **you already know the answer**.

...

Step 3 in action. Five orders with values you can add in your head, and the AI’s suggested line for the Nord average:

```
import pandas as pd

check = pd.DataFrame({
    "zone":      ["Nord", "Sued", "Nord", "Hafen", "Nord"],
    "total_eur": [10.0, 7.5, 20.0, 5.0, 30.0],
})
print(check[check["zone"] == "Nord"]["total_eur"].mean())
```

```
20.0
```

...

The Nord orders are 10, 20 and 30. You can average those in your head. Does the AI's answer match? Then the line has earned some trust on eight hundred rows. Verification is the job now, not typing.

Predict: does `.summarize()` exist?

The AI wrote this for Kevin. `orders` is a tiny two-row frame. What happens on the last line?

```
import pandas as pd
orders = pd.DataFrame({"zone": ["Nord", "Sued"], "total_eur": [12.0, 9.5]})

orders.summarize()
```

a) prints a summary table b) raises an `AttributeError` c) returns an empty DataFrame

...

Predict first. Commit to an answer before the next slide.

Answer: the method never existed

b) `AttributeError` — pandas has no `.summarize()`. The AI invented a plausible-sounding name, and Python answers with a traceback you've read since Episode 5:

```
import pandas as pd
orders = pd.DataFrame({"zone": ["Nord", "Sued"], "total_eur": [12.0, 9.5]})

orders.summarize() # the method the AI invented
```

```
AttributeError: 'DataFrame' object has no attribute 'summarize'
[31m-----[?]
[39m
[31mAttributeError[39m                                Traceback (most recent
call last)
[32m/var/folders/_5/jkkjxxdd5f1955l380dky7n80000gn/T/
ipykernel_66430/1876211805.py[39m in [36m?[39m[34m()[39m
[32m                                1[39m    [38;5;28;01mimport[39;00m    pandas
[38;5;28;01mas[39;00m pd
[32m                                2[39m    orders = pd.DataFrame({"[33m"zone"[39m:
[33m"Nord"[39m, [33m"Sued"[39m], [33m"total_eur"[39m:
[32m12.0[39m, [32m9.5[39m})
[32m                                3[39m
[32m----> [39m[32m4[39m                                orders.summarize()
[38;5;66;03m# the method the AI invented[39;00m

[32m~/development/lectures/Introduction-to-Python/.venv/lib/python3.12/
site-packages/pandas/core/generic.py[39m in [36m?[39m[34m(self,
```

```

name)
    6314
    38;5;28;01mand 39;00m name
    38;5;28;01mnot 39;00m 38;5;28;01min 39;00m self._accessors
    32m 6315
    38;5;28;01mand 39;00m
    self._info_axis._can_hold_identifiers_and_holds_name(name)
    32m 6316
    32m 6317
    38;5;28;01mreturn 39;00m self[name]
    32m-> 39m 32m6318
    38;5;28;01mreturn 39;00m
    object.__getattr__(self, name)

AttributeError: 'DataFrame' object has no attribute 'summarize'

```

...

The real method is `.describe()`:

```

print(orders.describe())           # the method that actually exists

```

```

      total_eur
count    2.000000
mean    10.750000
std      1.767767
min      9.500000
25%     10.125000
50%     10.750000
75%     11.375000
max     12.000000

```

...

An AI that sounds sure is not the same as an API that exists.

Disclosure: one line, every time

From Session VI the course rule stands: every submission that used AI carries a **one-line note** saying what you used it for.

- “Used the chatbot to draft the `groupby` line; I checked the totals by hand.”
- Not a confession. A professional reflex.

...

It protects **you**: it separates the work you understand from the work you pasted, so when a reviewer (or the investor) asks “how does this line work?”, you’re never caught claiming something you can’t explain.

When NOT to reach for AI

Sometimes the fastest path is the one you already own. You built a whole muscle in Part I:

- A red traceback? **You’ve read those since Episode 5.** The last line names the error and points at the file, often faster than describing it to a chatbot.
- A one-line filter you’ve written ten times? Just write it.

...

💡 Tip

Use AI to draft the unfamiliar and to explain the confusing, not to dodge the thinking you’re perfectly able to do. The pilot flies; the co-pilot advises.

⚡ Your turn — 10 minutes

Open the exercise (scan the QR or type the link):

beyondsimulations.github.io/Introduction-to-Python/notebooks/ex_08_a/



First **predict** what happens, then run it.

The DataFrame

What pandas adds over NumPy

Last session, a NumPy array turned a thousand numbers into one fast object, but every value had to share **one type**, and columns had no names.

- A **DataFrame** is a table: **named columns**, and each column can be a **different type**
- Text zones next to float euros next to integer counts, all in one object
- It’s the spreadsheet the investor handed you, loaded into Python

...

The convention everyone uses: `import pandas as pd`.

A DataFrame from a dictionary

Keys become **column names**; each list becomes a **column**. One order per row:

```
import pandas as pd

df = pd.DataFrame({
    "order_id": [101, 102, 103, 104, 105],
    "zone":     ["Nord", "Sued", "Nord", "Hafen", "Sued"],
    "items":    [2, 1, 3, 1, 4],
    "total_eur": [18.50, 7.20, 24.00, 6.80, 31.40],
})
print(df)
```

	order_id	zone	items	total_eur
0	101	Nord	2	18.5
1	102	Sued	1	7.2
2	103	Nord	3	24.0
3	104	Hafen	1	6.8
4	105	Sued	4	31.4

...

Five orders, four columns, mixed types (text, integers, floats) living happily together.

...

Note

You won't type the investor's eighty orders by hand. In tonight's lab a real CSV loads in one line, `orders = pd.read_csv("public/orders.csv")`. In the browser notebook that same line fetches the file over the web instead of from disk; pandas doesn't care, your code doesn't change.

First look: `.head()` and `.info()`

Before analyzing a table, glance at it. `.head()` shows the top rows; `.info()` reports columns, types, and counts:

```
print(df.head(3))    # first 3 rows
print()
df.info()            # columns, dtypes, non-null counts
```

	order_id	zone	items	total_eur
0	101	Nord	2	18.5
1	102	Sued	1	7.2
2	103	Nord	3	24.0

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   order_id    5 non-null      int64
1   zone        5 non-null      object
2   items       5 non-null      int64
3   total_eur   5 non-null      float64
dtypes: float64(1), int64(2), object(1)
memory usage: 292.0+ bytes
```

...

`.info()` is your first sanity check: right number of rows? Any column a surprising type?

The numbers at a glance: `.describe()`

`.describe()` computes count, mean, min, max and quartiles for every **numeric** column at once:

```
print(df.describe())
```

	order_id	items	total_eur
count	5.000000	5.000000	5.000000
mean	103.000000	2.200000	17.580000
std	1.581139	1.30384	10.688873
min	101.000000	1.000000	6.800000
25%	102.000000	1.000000	7.200000
50%	103.000000	2.000000	18.500000
75%	104.000000	3.000000	24.000000
max	105.000000	4.000000	31.400000

...

One call, the whole numeric summary: this is the `.summarize()` the AI wished existed, spelled correctly.

Selecting a column, filtering rows

Pick one column by name; keep rows with a **boolean mask**, exactly the NumPy idea from last session:

```
print(df["zone"])          # one column (a Series)
print()
print(df[df["zone"] == "Nord"])  # only the Nord rows - case-sensitive!
```

0	Nord
1	Sued

```
2    Nord
3    Hafen
4    Sued
Name: zone, dtype: object
```

```
   order_id  zone  items  total_eur
0        101  Nord     2        18.5
2        103  Nord     3        24.0
```

...

`df["zone"] == "Nord"` builds a column of `True/False`; indexing with it keeps the `True` rows. `.loc` exists for label-based selection, but a plain mask covers today.

Adding a column, sorting

Compute a new column from existing ones, then sort. Work on a **copy** so the original stays intact:

```
priced = df.copy()
priced["eur_per_item"] = priced["total_eur"] / priced["items"]
print(priced.sort_values("total_eur", ascending=False))
```

```
   order_id  zone  items  total_eur  eur_per_item
4        105  Sued     4        31.4          7.85
2        103  Nord     3        24.0          8.00
0        101  Nord     2        18.5          9.25
1        102  Sued     1         7.2          7.20
3        104  Hafen     1         6.8          6.80
```

...

`sort_values` reorders rows by a column, biggest orders first here. The new column is just arithmetic on two others, computed for every row at once.

Predict: filtering with the wrong case

Kevin filters for the Nord zone, but types it **lowercase**. The data spells it `"Nord"`. What does this print?

```
print(df[df["zone"] == "nord"])
```

a) an empty table, no error b) the Nord rows anyway: case is ignored c) a `KeyError` for the missing zone

...

Predict first. Commit to an answer before the next slide.

Answer: an empty table, no warning

a) an empty table — "nord" matches nothing, so the mask is all `False` and pandas hands back **zero rows**. No error, no complaint:

```
print(df[df["zone"] == "nord"])          # nothing matches "nord"
print("rows:", len(df[df["zone"] == "nord"]))
```

```
Empty DataFrame
Columns: [order_id, zone, items, total_eur]
Index: []
rows: 0
```

...

This is the dangerous kind of bug: it **runs**, it just returns nothing. **pandas won't warn you; YOU verify**, which is exactly why you test filters on a case whose answer you know.

One more tool: `groupby`

The investor's real question isn't about one row. It's **per zone**: which area brings in the most?

- `groupby` splits the table by a category, then computes **once per group**
- `df.groupby("zone")["total_eur"].sum()` → one total for each zone

...

Note

That's the teaser: tonight's lab does the heavy lifting with `groupby`, turning eighty raw orders into the handful of numbers the investor actually asked for.

Your turn — 10 minutes

Open the exercise (scan the QR or type the link):

beyondsimulations.github.io/Introduction-to-Python/notebooks/ex_08_b/



First **predict** what happens, then run it.

To the Lab

Tonight's episode

- Head to the lab notebook: [Episode 8 — The Data Room](#)
- The investor slides a **USB stick** across the table: every order, two full weeks. It's now `orders.csv`, eighty rows, loaded with one `pd.read_csv` line
- You'll `.head()`, `.info()` and `.describe()` it, filter with masks, add a column on a safe copy, and answer per-zone questions with `groupby`
- Kevin has discovered AI; your real job is to **supervise** it and catch the confident nonsense
- It runs entirely in your browser: no setup, just click and code

...

! Important

Download your `.py` before you leave. Closing the tab without downloading loses your work, the same motion you used to hand in the checkpoint this morning.

Wrap-up

Three things to remember

1. **AI is a co-pilot you verify.** Give it **context and constraints**, then **read it** → **run it** → **test it** on a case you know. An AI that sounds sure is not the same as an API that exists.
2. **A DataFrame is a named, mixed-type table.** `pd.DataFrame` from a dict, `pd.read_csv` from a file; `.head()`, `.info()`, `.describe()` to look before you leap.

3. **Select, filter, add, group.** `df["col"]`, a boolean mask (`df[df["zone"] == "Nord"]`, case-sensitive!), a new column on a `.copy()`, and `groupby` for per-category answers. pandas won't warn you; you verify.

...

Note

Next episode: charts the investor can't argue with. Numbers convince the careful; a good plot convinces the room. We turn the data room into pictures.

Literature

Books to start with

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. [Link to free online version](#)
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

...

Note

Working with AI this session? Revisit the [AI Tools page](#): context-and-constraints prompting, the verify workflow, and the one-line disclosure habit. For pandas itself, the official [10 minutes to pandas](#) guide is the friendliest next step.

...

For more, see the [literature list](#) of this course.